# Combined Science: Electrochemistry

IGCSE Worked Solutions

Each answer below is written using only the concepts taught in this chapter. If a question needs material beyond the chapter, it is flagged for teacher review.

### *Question 1.* *[Combined Science 2025 11]*

The diagram shows electrolysis apparatus. Which label shows the cathode?

Final answer: B

**1.** An electrolysis apparatus has three parts you can see in the diagram: the battery (DC source), the electrolyte (the liquid being split), and two electrodes dipped into the electrolyte. Only the electrodes can be the cathode; the battery and the electrolyte are not electrodes. [from: Electrolytic cells and electrodes]

**2.** Eliminate A and C: label A is the battery/cell and label C is the electrolyte, so neither can be the cathode. The cathode has to be one of the two electrodes, i.e. label B or label D. [from: Electrolytic cells and electrodes]

**3.** The cathode is defined as the negative electrode — it is the electrode connected to the negative terminal of the battery. The anode is the positive electrode, connected to the positive terminal. [from: Cathode: the negative electrode in an electrolytic cell; site of reduction; cations are discharged here.]

**4.** Label B is the electrode connected to the negative terminal, so B is the cathode. Label D is the electrode connected to the positive terminal, so D is the anode (not the cathode). [from: Anode: the positive electrode in an electrolytic cell; site of oxidation; anions are discharged here.]

From your chapter: Electrolytic cells and electrodes; Cathode: the negative electrode in an electrolytic cell; site of reduction; cations are discharged here.; Anode: the positive electrode in an electrolytic cell; site of oxidation; anions are discharged here.

### *Question 2.* *[Combined Science 2025 12]*

Which statement describes what happens during reduction?

Final answer: C — A compound loses oxygen.

**1.** The question asks what happens during reduction. In this chapter, reduction is introduced at the cathode of an electrolytic cell: positive ions (cations) move to the cathode and gain electrons. The chapter names this electron gain 'reduction'. [from: Movement of ions and charge transfer]

**2.** At IGCSE, reduction also has an equivalent everyday definition used when oxygen is involved: a compound is reduced when it LOSES oxygen. (For example, a metal oxide being reduced to the metal.) Gaining electrons and losing oxygen describe the same chemical change from different angles, so the chapter's idea of 'reduction at the cathode' matches the option that says the compound loses oxygen. [from: reduction at cathode]

**3.** Now check each option: (A) 'An element gains oxygen' — gaining oxygen is OXIDATION, not reduction, so A is wrong. (B) 'A compound gains oxygen' — this is also oxidation, so B is wrong. (C) 'A compound loses oxygen' — this matches the definition of reduction (the opposite of gaining oxygen). (D) 'A mixture of elements loses oxygen' — reduction is defined on a compound (an oxygen-containing substance), not on a mixture, so D is wrong. [from: oxidation at anode]

**4.** Therefore the correct choice is C: a compound loses oxygen. [from: Movement of ions and charge transfer]

From your chapter: Movement of ions and charge transfer; reduction at cathode; oxidation at anode

### *Question 3.* *[Combined Science 2025 11]*

Which statement explains why aluminium is extracted from its ore by electrolysis and not by heating with carbon?

Final answer: A

**1.** Read the question. It asks for the reason aluminium is extracted from its ore by electrolysis rather than by heating the ore with carbon. [from: Electrolysis of molten binary ionic compounds]

**2.** From the chapter, electrolysis of a molten binary ionic compound produces the metal at the cathode and the non-metal at the anode, and this method is used industrially to extract reactive metals such as sodium, magnesium and aluminium. [from: Electrolysis of molten binary ionic compounds]

**3.** The chapter contrasts this with iron, which is less reactive and is extracted by heating its ore with carbon in a blast furnace, NOT by electrolysis. So the deciding factor is whether the metal is more reactive than carbon. [from: Electrolysis of molten binary ionic compounds]

**4.** Check each option against this rule. A: 'Aluminium is more reactive than carbon' matches the rule exactly — carbon cannot pull the oxygen away from aluminium oxide, so electrolysis must be used. B: comparing to hydrogen is irrelevant because the alternative method uses carbon, not hydrogen. C: a melting point of 660 °C is a physical property that would not stop carbon reduction. D: corrosion resistance is a property of the finished metal, not a reason for choosing an extraction method. [from: Electrolysis of molten binary ionic compounds]

**5.** Therefore the statement that explains the choice of electrolysis is A. [from: Electrolysis of molten binary ionic compounds]

From your chapter: Electrolysis of molten binary ionic compounds

### *Question 4.* *[Combined Science 2025 13]*

The diagram shows the apparatus for electrolysis with two electrodes labelled W (left, connected to the negative terminal) and X (right, connected to the positive terminal) dipping into a solution labelled Y. Which row identifies W, X and Y? (Columns: W, X, Y.)

Final answer: C (W = cathode, X = anode, Y = electrolyte)

**1.** Identify W: W is connected to the negative terminal of the battery. In an electrolytic cell, the electrode connected to the negative terminal is called the cathode. So W = cathode. [from: Electrolytic cells and electrodes]

**2.** Identify X: X is connected to the positive terminal of the battery. The electrode connected to the positive terminal is called the anode. So X = anode. [from: Electrolytic cells and electrodes]

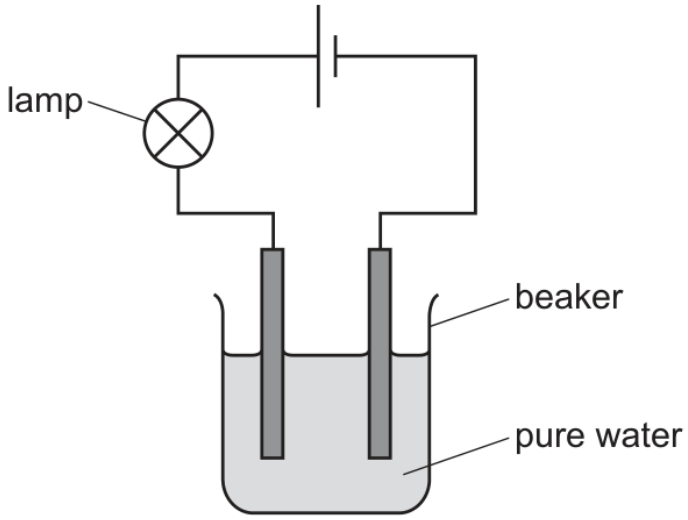
**3.** Identify Y: Y is the liquid that the two electrodes dip into. The liquid that conducts electricity by ion movement and is decomposed in the process is called the electrolyte. So Y = electrolyte. (It cannot be the power supply, because the power supply is the battery providing the DC current, not the liquid between the electrodes.) [from: Electrolyte]

**4.** Match W = cathode, X = anode, Y = electrolyte to the options. Option A has W and X swapped. Options B and D say Y is the power supply, which is wrong. Only option C matches: cathode; anode; electrolyte. [from: Electrolytic cells and electrodes]

From your chapter: Electrolytic cells and electrodes; Cathode: the negative electrode in an electrolytic cell; site of reduction; cations are discharged here.; Anode: the positive electrode in an electrolytic cell; site of oxidation; anions are discharged here.; Electrolyte: a liquid (molten ionic compound or aqueous solution of ions) that conducts electricity by the movement of ions and is decomposed in the process.

***Question 5.*** *[Combined Science 2025 22]*

**19** The apparatus shown is used to test a property of compound R.




The lamp does **not** light when the beaker contains pure water.

When compound R is dissolved in the water, the lamp lights.

Which row describes compound R?

| | type of bonding | elements of compound R |
|---|---|---|
| **A** | covalent | a metal and a non-metal |
| **B** | covalent | non-metals only |
| **C** | ionic | non-metals only |
| **D** | ionic | a metal and a non-metal |

Final answer: D

**1.** Pure water does not light the lamp, so pure water is a non-electrolyte (it has no free-moving charged particles to carry the current). [from: Conductivity and electrolytes]

**2.** When compound R dissolves in water the lamp lights, which means the solution now conducts electricity. A liquid that conducts electricity by the movement of ions is called an electrolyte. [from: Electrolyte: a liquid (molten ionic compound or aqueous solution of ions) that conducts electricity by the movement of ions and is decomposed in the process.]

**3.** For a solution to conduct, the dissolved compound must split up into ions that are free to move. Covalent compounds dissolve as neutral molecules, so they do not make the solution conduct. Only ionic compounds release ions into solution, so R must be ionic. [from: Conductivity and electrolytes]

**4.** Ionic bonding is the bonding between a metal (which loses electrons to form a positive ion) and a non-metal (which gains electrons to form a negative ion). So an ionic compound is made of a metal combined with a non-metal. [from: ionic bonding]

**5.** Compound R must therefore be ionic and made of a metal and a non-metal. That matches option D. Option C (ionic, non-metals only) is impossible because ionic compounds are not made of non-metals only. Options A and B are covalent and would not conduct. [from: Conductivity and electrolytes]

From your chapter: Conductivity and electrolytes; Electrolyte: a liquid (molten ionic compound or aqueous solution of ions) that conducts electricity by the movement of ions and is decomposed in the process.; ionic bonding

### *Question 6.* *[Combined Science 2025 12]*

**18** What is produced at the anode during the electrolysis

**A** bromide ions

**B** bromine

**C** lead

**D** lead(II) ions

Final answer: B (bromine)

**1.** Molten lead(II) bromide, PbBr2(l), is a molten binary ionic compound. When it melts, the ions become free to move, so it behaves as an electrolyte and can be split up by electricity. [from: Electrolysis of molten binary ionic compounds]

**2.** In the liquid, PbBr2 contains Pb^2+ cations (positive) and Br- anions (negative). Anions are attracted to the anode (the positive electrode), so the Br- ions move to the anode. [from: Movement of ions and charge transfer]

**3.** At the anode, the Br- ions each lose an electron (this is oxidation). Two Br- ions join together to form one molecule of bromine gas. Half-equation: 2Br- -> Br2 + 2e-. [from: 2Br- -> Br2 + 2e-]

**4.** So at the anode the product formed is the non-metal bromine (Br2), not the bromide ion (which is the ion before it loses its electron), and not lead or lead(II) ions (those belong at the cathode). The correct option is B: bromine. [from: Electrolysis of molten binary ionic compounds]

From your chapter: Electrolysis of molten binary ionic compounds; Movement of ions and charge transfer; Ionic half-equations; 2Br- -> Br2 + 2e-; PbBr2(l) -> Pb(l) + Br2(g)

### *Question 7.* *[Combined Science 2025 23]*

Apparatus used to electrolyse molten lead(II) bromide is shown (d.c. power supply connected to two electrodes in molten lead(II) bromide; electrodes labelled P at top, Q at bottom/electrolyte). Which row identifies P and Q?

Final answer: C

**1.** An electrolytic cell has three essential parts: a d.c. power supply (battery), two electrodes dipped into a liquid, and the liquid itself, which is the electrolyte. Identifying P and Q means labelling which part each letter points to in the diagram. [from: Electrolytic cells and electrodes]

**2.** The liquid being electrolysed here is molten lead(II) bromide. Because its ions (Pb^2+ and Br-) are free to move when molten, it conducts electricity and is therefore an electrolyte, not an electrode (electrodes are the solid graphite/platinum rods dipping into it). [from: Conductivity and electrolytes]

**3.** Q is the label placed on the molten lead(II) bromide at the bottom of the container, so Q must be the electrolyte. This rules out options B and D, which call Q an electrode. [from: Electrolyte: a liquid (molten ionic compound or aqueous solution of ions) that conducts electricity by the movement of ions and is decomposed in the process.]

**4.** The two electrodes are wired to the d.c. power supply: one to the positive terminal (the anode) and one to the negative terminal (the cathode). P labels the electrode connected to the positive terminal of the power supply, so P is the anode (positive electrode). [from: Electrolytic cells and electrodes]

**5.** Combining both labels: P = anode and Q = electrolyte. This matches option C. [from: Electrolytic cells and electrodes]

From your chapter: Electrolytic cells and electrodes; Conductivity and electrolytes; Electrolyte: a liquid (molten ionic compound or aqueous solution of ions) that conducts electricity by the movement of ions and is decomposed in the process.; Anode: the positive electrode in an electrolytic cell; site of oxidation; anions are discharged here.

## *Question 8.* *[Combined Science 2025 32]*

1 **(a)** Fig. 1.1 shows a white blood cell.

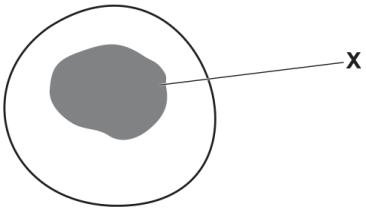



**Fig. 1.1**

**(i)** Name the part labelled **X** on Fig. 1.1.

..... [1]

**(ii)** State **one** function of white blood cells.

..... [1]

**(iii)** Circle **two** structures found in both white blood cells and bacterial cells.

| | | |
|---|---|---|
| **cell membrane** | **cell wall** | **cytoplasm** |
| **circular DNA** | **large vacuole** | **plasmid** |

[2]

**(b)** Blood is part of the circulatory system.

Complete the sentence to describe the circulatory system.

The circulatory system is a system of blood ..... with a pump

and ..... to ensure one–way flow of blood.

[2]



Final answer: (d) reduction; (e) bauxite; (f)(i) electrolysis: sodium (or magnesium); heating with carbon: zin

**1.** (d) In the reaction $Fe_2O_3 + 3CO \rightarrow 2Fe + 3CO_2$, the iron(III) oxide ($Fe_2O_3$) loses its oxygen to become iron (Fe). Losing oxygen is called reduction, so we circle 'reduction'. (This is the same word used in the chapter for 'reduction at the cathode' — gaining electrons and/or

losing oxygen.) [from: out_of_scope_notes: Extraction of iron is by reduction in a blast furnace, NOT electrolysis]

**2.** (e) The chapter explains that aluminium is extracted by electrolysis of its molten ionic compound (mixed with cryolite to lower the melting point). The ore that this ionic compound comes from is called bauxite. [from: Electrolysis of molten binary ionic compounds]

**3.** (f)(i) The chapter says electrolysis is used industrially to extract the MORE reactive metals: sodium, magnesium and aluminium. Looking at the reactivity series given (sodium, magnesium, carbon, zinc, hydrogen, copper), pick a metal ABOVE carbon for electrolysis: sodium (or magnesium). Pick a metal BELOW carbon for heating with carbon: zinc (or copper). [from: Electrolysis of molten binary ionic compounds]

**4.** (f)(ii) Carbon can only pull the oxygen away from a metal oxide if carbon is MORE reactive than that metal. Sodium is more reactive than carbon, so carbon cannot reduce sodium compounds — the only way to get sodium out is to electrolyse its molten ionic compound (the chapter's method for reactive metals Na, Mg, Al). Zinc is less reactive than carbon, so carbon CAN reduce zinc oxide; that is why heating with carbon works for zinc but not for sodium. [from: out_of_scope_notes: electrolysis is used for extracting more reactive metals such as aluminium, sodium, magnesium]

From your chapter: Electrolysis of molten binary ionic compounds; out_of_scope_notes: Extraction of iron is by reduction in a blast furnace, NOT electrolysis. The chapter only mentions electrolysis is used for extracting more reactive metals such as aluminium, sodium, magnesium.; Electrolysis: the breakdown of an ionic compound, when molten or in solution, by the passage of an electric current.

### *Question 9.* *[Combined Science 2025 21]*

Which row shows what forms at the electrodes during electrolysis of an aqueous compound? Columns: anode, cathode.

Final answer: B

**1.** During electrolysis of an aqueous (dissolved) compound, positive ions (cations) move to the cathode (the negative electrode) and negative ions (anions) move to the anode (the positive electrode). Cations gain electrons (reduction) at the cathode; anions lose electrons (oxidation) at the anode. [from: Movement of ions and charge transfer]

**2.** At the anode of an aqueous electrolyte, the discharged ion is always a non-metal. Examples from the chapter: in concentrated NaCl(aq) chlorine (Cl2) is produced at the anode; in dilute H2SO4 oxygen (O2) is produced at the anode; in CuSO4(aq) with inert electrodes oxygen is produced at the anode. In every case the anode product is a non-metal. [from: Electrolysis of concentrated sodium chloride solution]

**3.** At the cathode of an aqueous electrolyte, a metal CAN be deposited. The chapter gives CuSO4(aq): copper (a metal) is formed at the cathode (Cu^2+ + 2e- -> Cu). So 'cathode: metal' is a valid cathode product for an aqueous compound. [from: Electrolysis of aqueous copper(II) sulfate]

**4.** Check each option against the rules above. A (anode metal, cathode non-metal) is wrong because metals never form at the anode and non-metals do not form at the cathode from cations. C (anode hydrogen) is wrong because hydrogen is a cation (H+) and is discharged at the cathode, not the anode. D (anode hydrogen, cathode oxygen) is wrong for the same reason; also O^2- / OH- go to the anode, not the cathode. B (anode non-metal, cathode metal) matches the pattern shown by CuSO4(aq): oxygen (non-metal) at anode, copper (metal) at cathode. [from: Movement of ions and charge transfer]

**5.** Therefore the correct row is B: anode = non-metal, cathode = metal. [from: Electrolysis of aqueous copper(II) sulfate]

From your chapter: Movement of ions and charge transfer; Electrolysis of concentrated sodium chloride solution; Electrolysis of aqueous copper(II) sulfate; Electrolytic cells and electrodes

## *Question 10.* *[Combined Science 2025 43]*

**4** Table 4.1 shows information about three compounds, **A**, **B** and **C**.

**Table 4.1**

| compound | melting point / °C | boiling point / °C | electrical conductivity |
|---|---|---|---|
| **A** | 801 | 1465 | conducts only when molten or in aqueous solution |
| **B** | 1713 | 2950 | does **not** conduct |
| **C** | –210 | –196 | does **not** conduct |

**(a)** **(i)** Identify the compound in Table 4.1 that is a solid at 25 °C **and** has only covalent bonding.

Explain your choice.

compound .....

explanation .....

.....

.....

[2]

**(ii)** Identify the compound in Table 4.1 that is a simple molecular compound.

Explain your choice.

compound .....

explanation .....

.....

.....

**(b)** The temperature of compound **A** is increased from 800 °C to 802 °C.

Describe how the arrangement, energy and motion of the particles in compound **A** change with this increase in temperature.

arrangement .....

.....

energy .....

.....

motion .....

.....
[3]

**(c)** An electric current is passed through an aqueous solution of compound **A**.

A chemical change occurs.

Explain why this is a chemical change.

.....

.....

..... [2]

[Total: 8]

Final answer: (b) Arrangement: ions change from a regular, fixed ionic lattice (solid) to a random, disordere

**1.** (b) At 800 C compound A is a solid ionic compound (its melting point is about 801 C). In the solid state the positive and negative ions are held together by ionic bonding in a regular, fixed lattice: they cannot move out of their positions. This is why solid ionic compounds are non-electrolytes. [from: Conductivity and electrolytes]

**2.** (b) As the temperature rises past the melting point, compound A melts. The ions gain enough energy to break away from the lattice. So between 800 C and 802 C the ions' energy increases. [from: Conductivity and electrolytes (ionic compounds only conduct when molten because the ions become free to move)]

**3.** (b) At 802 C compound A is molten. The arrangement is now random/disordered (ions are still close together but no longer in a fixed lattice) and the ions are free to move throughout the liquid. This free movement of ions is what makes the molten compound an electrolyte. [from: Electrolysis of molten binary ionic compounds]

**4.** (c) Passing an electric current through an aqueous solution of compound A is electrolysis. Electrolysis is defined as the breakdown of an ionic compound, when molten or in solution, by the passage of an electric current. [from: Electrolysis: the breakdown of an ionic compound, when molten or in solution, by the passage of an electric current.]

**5.** (c) It is a chemical change because new substances are formed: at the cathode (negative electrode) cations gain electrons and are discharged as a new product, and at the anode (positive electrode) anions lose electrons and are discharged as a new product. These products are different from the compound A that was dissolved in the water, so a chemical change has taken place. [from: Movement of ions and charge transfer]

From your chapter: Conductivity and electrolytes; Electrolysis of molten binary ionic compounds; Movement of ions and charge transfer; Electrolysis: the breakdown of an ionic compound, when molten or in solution, by the passage of an electric current.

### *Question 11.* *[Combined Science 2025 42]*

Continuation of Question 4 on sodium fluoride, now addressing the electrolysis of molten NaF (Fig. 4.2 shows inert electrodes dipped in molten sodium fluoride with a power supply) and the reaction of aqueous NaF with aqueous silver nitrate.

Final answer: c(i) Electrolysis is the breakdown of an ionic compound, when molten or in solution, by the pa

**1.** (c)(i) State the chapter definition of electrolysis: the breakdown of an ionic compound, when molten or in solution, by the passage of an electric current. Two mark-bearing ideas: (a) breakdown of an ionic compound when molten or in solution, (b) by passing an electric current through it. [from: Electrolysis: the breakdown of an ionic compound, when molten or in solution, by the passage of an electric current.]

**2.** (c)(ii) Identify molten NaF as a molten binary ionic compound (it contains only two elements, Na+ and F- ions, and is melted so the ions are free to move and carry charge). [from: Electrolysis of molten binary ionic compounds]

**3.** (c)(ii) Apply the rule for molten binary ionic compounds: the metal is always formed at the cathode (negative electrode) and the non-metal is always formed at the anode (positive electrode). So Na forms at the negative electrode and F2 forms at the positive electrode. [from: Electrolysis of molten binary ionic compounds]

**4.** (c)(ii) Confirm using ion movement: Na+ (cation) moves to the cathode and gains electrons to become Na metal (reduction). F- (anion) moves to the anode and loses electrons, pairing up to become F2 gas (oxidation). [from: Movement of ions and charge transfer]

**5.** (d) Use the word equation: aqueous sodium fluoride + aqueous silver nitrate -> aqueous sodium nitrate + one other product. The 'other product' is silver fluoride, AgF (Ag+ combines with F- in a 1:1 ratio because both ions carry a single charge, giving a neutral binary compound). [from: Binary compound]

**6.** (d) Write and balance the symbol equation. All four species have 1 Na, 1 Ag, 1 F, and 1 NO3, so coefficients of 1 on every term give a balanced equation: NaF(aq) + AgNO3(aq) -> NaNO3(aq) + AgF. [from: Binary compound]

From your chapter: Electrolysis: the breakdown of an ionic compound, when molten or in solution, by the passage of an electric current.; Electrolysis of molten binary ionic compounds; Movement of ions and charge transfer; Binary compound

### *Question 12.* *[Combined Science 2025 22]*

**27** Which row shows an electrolyte and the products of electrolysis for this electrolyte?

| | electrolyte | products |
|---|---|---|
| **A** | concentrated aqueous sodium chloride | sodium and chlorine |
| **B** | molten potassium bromide | hydrogen and bromine |
| **C** | molten sodium chloride | sodium and chlorine |
| **D** | dilute sulfuric acid | hydrogen and sulfur dioxide |

Final answer: C

**1.** Check each row by applying the chapter's electrolysis rules for the stated electrolyte (molten binary ionic compound vs aqueous solution vs dilute acid). [from: Movement of ions and charge transfer]

**2.** A: concentrated aqueous NaCl contains Na+, H+, Cl-, OH-. H+ is preferentially discharged at the cathode, so hydrogen (not sodium) is produced; chlorine is produced at the anode. Predicted products = hydrogen + chlorine, not sodium + chlorine. Row A is wrong. [from: Electrolysis of concentrated sodium chloride solution]

**3.** B: molten potassium bromide is a molten binary ionic compound, so the metal (potassium) forms at the cathode and the non-metal (bromine) forms at the anode. Hydrogen is not produced because there are no H+ ions in a molten salt. Predicted products = potassium + bromine, not hydrogen + bromine. Row B is wrong. [from: Electrolysis of molten binary ionic compounds]

**4.** C: molten sodium chloride is a molten binary ionic compound (Na+ and Cl-). The metal (sodium) is discharged at the cathode and the non-metal (chlorine) at the anode. Predicted products = sodium + chlorine. Row C matches. [from: Electrolysis of molten binary ionic compounds]

**5.** D: electrolysing dilute sulfuric acid effectively electrolyses water, giving hydrogen at the cathode and oxygen at the anode (2:1 by volume). No sulfur dioxide is produced. Predicted products = hydrogen + oxygen, not hydrogen + sulfur dioxide. Row D is wrong. [from: Electrolysis of dilute sulfuric acid]

**6.** Only row C correctly pairs an electrolyte with its real electrolysis products, so the answer is C. [from: Electrolysis of molten binary ionic compounds]

From your chapter: Movement of ions and charge transfer; Electrolysis of molten binary ionic compounds; Electrolysis of concentrated sodium chloride solution; Electrolysis of dilute sulfuric acid; $PbBr_2(l) \rightarrow Pb(l) + Br_2(g)$; $2H_2O(l) \rightarrow 2H_2(g) + O_2(g)$

## *Question 13.* *[Combined Science 2025 42]*

**(c)** The reaction between aluminium and iron(III) oxide is exothermic.

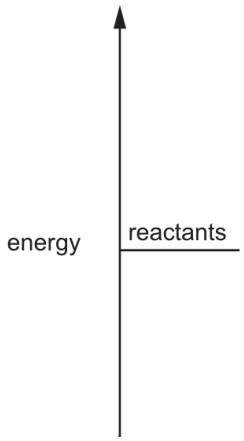
A flame is used to provide the activation energy to start the reaction.

Fig. 5.1 shows the incomplete reaction pathway diagram for this reaction.

Complete Fig. 5.1.

Include:

- the energy level of the products
- a labelled arrow to show the activation energy, $E_a$
- a labelled arrow to show the overall energy change of the reaction.



**5** The reaction between aluminium and iron(III) oxide is used to join iron railway lines together.

$$2Al(s) + Fe_2O_3(s) \rightarrow 2Fe(l) + Al_2O_3(s)$$

Molten iron is produced in the reaction. Molten iron becomes solid when it cools.

**(a)** **(i)** State how the equation shows that **molten** iron is produced.

..... [1]

**(ii)** Use the equation to explain why this reaction is a redox reaction.

.....

.....

..... [2]

**(iii)** Complete the ionic equation for this reaction.

$Al + Fe^{3+} \rightarrow$ ..... + ..... [2]

**(b)** Explain the meaning of the symbol (III) in iron(III) oxide.

.....

..... [1]

Final answer: (a)(i) The state symbol (l) next to Fe in the equation means Fe is a liquid (molten). (a)(ii) Al lo

**1.** (a)(i) In chemical equations we write a state symbol in brackets after each formula: (s) = solid, (l) = liquid/molten, (g) = gas, (aq) = dissolved in water. The equation shows Fe(l), so the state symbol (l) tells us the iron is produced as a liquid, i.e. molten. [from: PbBr2(l) -> Pb(l) + Br2(g)]

**2.** (a)(ii) Oxidation means loss of electrons, reduction means gain of electrons. A reaction that contains both is called a redox reaction. In the thermite reaction the aluminium atom Al has no charge, but in Al2O3 the aluminium is present as Al^3+, so each Al atom has lost 3 electrons (oxidised). [from: Movement of ions and charge transfer]

**3.** (a)(ii) In Fe2O3 the iron is present as Fe^3+ (charge of 3+). In the product the iron atoms have no charge, so each Fe^3+ ion has gained 3 electrons (reduced). Because oxidation (of Al) and reduction (of Fe^3+) both happen in the same reaction, this is a redox reaction. [from: Ionic half-equations]

**4.** (a)(iii) Write the two half-equations using the chapter pattern for electron transfer. Oxidation (loss of electrons): Al -> Al^3+ + 3e-. Reduction (gain of electrons): Fe^3+ + 3e- -> Fe. [from: Ionic half-equations]

**5.** (a)(iii) Add the two half-equations. The 3 electrons on each side cancel, giving the balanced ionic equation: Al + Fe^3+ -> Al^3+ + Fe. The numbers of each atom and the total charge (+3) match on both sides, so it is correctly balanced. [from: Ionic half-equations]

**6.** (b) Iron can form ions with more than one possible charge (e.g. Fe^2+ or Fe^3+). The Roman numeral (III) in the name iron(III) oxide tells us which one is present: each iron ion carries a charge of 3+, so the compound contains Fe^3+ ions. This matches the question's ionic equation, which uses Fe^3+. [from: Cu^2+ + 2e- -> Cu]

From your chapter: Movement of ions and charge transfer; Ionic half-equations; PbBr2(l) -> Pb(l) + Br2(g); Cu^2+ + 2e- -> Cu; Cation: a positively charged ion that moves towards the cathode.

### *Question 14.* *[Combined Science 2025 33]*

Continuation of Question 4 on reaction between magnesium and steam.

Final answer: (a)(iii) electrons; cations; ionic; oxidation. (b) observation: no (visible) reaction / no change; e

**1.** (a)(iii) First blank — When magnesium atoms react, each atom LOSES two outer-shell negative particles to become a Mg2+ ion. In this chapter we call those negative particles 'electrons' (the same particles that flow through a wire during electrolysis). So the first blank = electrons. [from: Movement of ions and charge transfer (oxidation at anode = loss of electrons)]

**2.** (a)(iii) Second blank — A positively charged ion is called a cation. In the chapter's electrolysis work, cations (e.g. Pb2+, Cu2+, H+, Na+) move to the cathode. So Mg2+ is a cation. Second blank = cations. [from: Movement of ions and charge transfer / Definition: Cation = a positively charged ion that moves towards the cathode]

**3.** (a)(iii) Third blank — The chapter's 'Conductivity and electrolytes' topic tells us that positive ions and negative ions attract each other and hold together as an ionic compound. That attraction is an ionic bond. Third blank = ionic. [from: Conductivity and electrolytes (ionic bonding holds oppositely charged ions together)]

**4.** (a)(iii) Fourth blank — Losing electrons is the definition of oxidation in this chapter (it's what happens to anions at the anode: 2Br- -> Br2 + 2e-). 'Gaining oxygen' is the older, equivalent definition of the same process: both mean electrons have been lost. Magnesium atoms lose 2 electrons, so the reaction is oxidation. Fourth blank = oxidation. [from: Movement of ions and charge transfer (oxidation = loss of electrons at anode); Ionic half-equations (e.g. 2Br- -> Br2 + 2e-)]

**5.** (b) Out-of-chapter note — Part (b) asks us to PREDICT what happens when copper replaces magnesium with steam. Answering this correctly needs the reactivity series of metals (and the rule that metals below hydrogen, such as copper, do not react with water or steam). The Electrochemistry chapter profile does NOT list the reactivity series as a covered topic. From general IGCSE chemistry knowledge the expected answer is: observation = 'no reaction / no change'; explanation = 'copper is less reactive than magnesium and does not react with steam'. I have flagged this sub-part because the justifying concept is outside this chapter's scope. [from: out_of_scope: reactivity series / metal + water reactions (not in Electrochemistry chapter profile)]

From your chapter: Conductivity and electrolytes; Movement of ions and charge transfer; Ionic half-equations; Cation: a positively charged ion that moves towards the cathode

### *Question 15.* *[Combined Science 2025 41]*

**5** **(a)** Fig. 5.1 shows the electrolysis of molten zinc chloride, $ZnCl_2$, using inert electrodes.

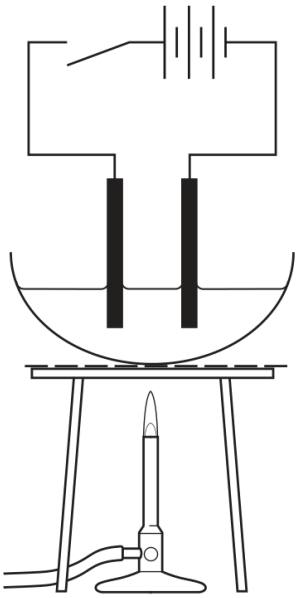


**Fig. 5.1**

**(i)** The switch is closed.

Describe the observation at the positive electrode.

.....

..... [1]

Final answer: (a)(i) A pale yellow-green gas (chlorine) is given off / bubbles form at the positive electrode. (

**1.** (a)(i) The positive electrode is the anode. Molten ZnCl2 is a binary ionic compound containing Zn^2+ and Cl- ions that are free to move. The negative Cl- ions (anions) are attracted to the anode. [from: Movement of ions and charge transfer]

**2.** (a)(i) At the anode the Cl- ions lose electrons (oxidation) to form chlorine atoms, which pair up to form Cl2 molecules. Chlorine is a pale yellow-green gas, so the observation is bubbles of a pale yellow-green gas given off at the positive electrode. [from: Electrolysis of molten binary ionic compounds]

**3.** (a)(ii) Zinc is formed at the cathode (negative electrode) because Zn^2+ is a positive ion (cation) and is attracted to the cathode, where it gains electrons (reduction). [from: Movement of ions and charge transfer]

**4.** (a)(ii) A Zn^2+ ion carries a 2+ charge, so it must gain 2 electrons to become a neutral Zn atom. This gives the ionic half-equation Zn^2+ + 2e- -> Zn. [from: Ionic half-equations]

**5.** (a)(ii) State symbols: the Zn^2+ ions come from molten zinc chloride so they are written as (l). The zinc metal formed at the cathode in the molten electrolyte is also written as (l), following the convention used for molten PbBr2 giving Pb(l). Final equation: Zn^2+(l) + 2e- -> Zn(l). [from: PbBr2(l) -> Pb(l) + Br2(g)]

From your chapter: Movement of ions and charge transfer; Electrolysis of molten binary ionic compounds; Ionic half-equations; PbBr2(l) -> Pb(l) + Br2(g); Cathode: the negative electrode in an electrolytic cell; site of reduction; cations are discharged here.; Anode: the positive electrode in an electrolytic cell; site of oxidation; anions are discharged here.

***Question 16.*** *[Combined Science 2025 12]*

**22** Which statement describes the extraction of aluminium?

**A** It is extracted from bauxite using electrolysis.

**B** It is extracted from bauxite using carbon.

**C** It is extracted from hematite using electrolysis.

**D** It is extracted from hematite using carbon.

Final answer: A

**1.** Aluminium is a reactive metal, so it cannot be extracted by heating its ore with carbon. The chapter states that electrolysis is used industrially for extracting reactive metals such as sodium, magnesium and aluminium. [from: Electrolysis of molten binary ionic compounds]

**2.** The ore of aluminium is bauxite (not hematite; hematite is an iron ore). Aluminium is extracted by electrolysing molten aluminium oxide (from bauxite) mixed with cryolite to lower the melting point. [from: aluminium extraction]

**3.** Therefore the correct statement is A: aluminium is extracted from bauxite using electrolysis. [from: industrial extraction]

From your chapter: Electrolysis of molten binary ionic compounds; aluminium extraction; cryolite; industrial extraction

***Question 17.*** *[Combined Science 2025 31]*

**(b)** Fig. 5.2 shows part of the Periodic Table.

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| 3<br>Li<br>lithium<br>7 | 4<br>Be<br>beryllium<br>9 | 5<br>B<br>boron<br>11 | 6<br>C<br>carbon<br>12 | 7<br>N<br>nitrogen<br>14 | 8<br>O<br>oxygen<br>16 | 9<br>F<br>fluorine<br>19 | 10<br>Ne<br>neon<br>20 |

**Fig. 5.2**

**(i)** Circle the word that describes the part of the Periodic Table shown in Fig. 5.2.

**group** **period** **property** **transition**

[1]

**(ii)** Identify an element in Fig. 5.2 that matches each description.

Each element may be used once, more than once or not at all.

- forms an ion with a charge of 1–

  .....

- has only two outer-shell electrons

  .....

- is used as an inert electrode in electrolysis

  .....

- reacts with oxygen to form a gas that leads to global warming

  .....

[4]

**(iii)** State how the metallic character of the elements shown in Fig. 5.2 changes from Li to Ne.

..... [1]

[Total: 8]

Final answer: b(ii) 'used as an inert electrode in electrolysis' = C (carbon; graphite is the form used). The o

**1.** (b)(i) This sub-part asks about the group / period / property / transition labels for the Periodic Table. That terminology belongs to the Periodic Table chapter, not Electrochemistry, so it is out of scope here — review with your teacher. [from: out_of_scope: Periodic Table terminology]

**2.** (b)(ii) descriptor 'forms an ion with a charge of 1-': working out the charge on an ion from its Periodic Table group is atomic-structure / Periodic Table content, so it is out of scope for this chapter — review with your teacher. [from: out_of_scope: ion formation from group number]

**3.** (b)(ii) descriptor 'has only two outer-shell electrons': electron-shell / electronic-configuration reasoning is atomic-structure content, not Electrochemistry, so it is out of scope — review with your teacher. [from: out_of_scope: electron configuration]

**4.** (b)(ii) descriptor 'is used as an inert electrode in electrolysis' (THIS ONE IS IN SCOPE): in this chapter, an inert electrode is defined as an electrode that does not react with the electrolyte or with the products. The chapter names graphite and platinum as the two

standard inert electrodes. [from: Inert electrode: an electrode that does not react with the electrolyte or the products (e.g. graphite, platinum).]

**5.** (b)(ii) continued: looking along Period 2 in Fig. 5.2 (Li, Be, B, C, N, O, F, Ne), the element that is used as an inert electrode (in the form of graphite) is carbon. So the answer to this descriptor is C. [from: Electrolytic cells and electrodes]

**6.** (b)(ii) descriptor 'reacts with oxygen to form a gas that leads to global warming': this is about combustion products and greenhouse gases, which sits in the Earth/atmosphere chapter, not Electrochemistry, so it is out of scope — review with your teacher. [from: out_of_scope: greenhouse gases / combustion]

**7.** (b)(iii) 'how metallic character changes from Li to Ne': this is a Periodic Table trend question, which is not covered in the Electrochemistry chapter, so it is out of scope — review with your teacher. [from: out_of_scope: Periodic Table trends]

From your chapter: Electrolytic cells and electrodes; Inert electrode: an electrode that does not react with the electrolyte or the products (e.g. graphite, platinum).

### *Question 18.* *[Combined Science 2025 33]*

Continuation of Question 4 on reaction between magnesium and steam.

Final answer: (a)(iii) Blank 1: electrons. Blank 2: cations. Blank 3: ionic. Blank 4: oxidation. (b) Observation

**1.** (a)(iii) Quick answers in order — Blank 1: electrons. Blank 2: cations. Blank 3: ionic. Blank 4: oxidation. The next steps show the reasoning for each blank. [from: Movement of ions and charge transfer]

**2.** (a)(iii) Blank 1 — 'magnesium atoms lose ____ to form positive ions'. An atom becomes a positive ion by losing negatively charged particles. From the chapter, electron transfer is how atoms and ions swap charge (e.g. Cu^2+ + 2e- -> Cu goes the other way when an atom turns into a cation). So Mg atoms lose ELECTRONS. [from: Movement of ions and charge transfer]

**3.** (a)(iii) Blank 2 — 'These positive ions are known as ____'. The chapter glossary says a cation is 'a positively charged ion that moves towards the cathode'. Mg^2+ is a positive ion, so it is a CATION. [from: Cation: a positively charged ion that moves towards the cathode.]

**4.** (a)(iii) Blank 3 — 'Positive ions and negative ions attract each other and form ____ bonds'. In the chapter's 'Conductivity and electrolytes' topic, ionic bonding is a key term: the pull between oppositely charged ions is an IONIC bond. [from: Conductivity and electrolytes]

**5.** (a)(iii) Blank 4 — 'magnesium atoms gain oxygen. This type of reaction is ____'. The chapter defines oxidation as loss of electrons at the anode (for example 2Br- -> Br2 + 2e-). Mg loses electrons when it bonds with oxygen, and gaining oxygen is also the classical meaning of the word. So the reaction is OXIDATION. [from: Ionic half-equations]

**6.** (b) Observation: no reaction — no bubbles of hydrogen gas are given off and no black solid (copper(II) oxide) forms on the copper. Explanation: copper is less reactive than magnesium, so copper cannot take oxygen from water (steam). Study tip: this chapter tells us that electrolysis is used for 'more reactive metals such as aluminium, sodium, magnesium' — copper is not in that group, which fits with copper being too unreactive to react with steam. (The full reactivity series is covered more deeply in the metals chapter.) [from: Extraction of iron is by reduction in a blast furnace, NOT electrolysis. The chapter only mentions electrolysis is used for extracting more reactive metals such as aluminium, sodium, magnesium.]

From your chapter: Movement of ions and charge transfer; Cation: a positively charged ion that moves towards the cathode.; Conductivity and electrolytes; Ionic half-equations; Extraction of iron is by reduction in a blast furnace, NOT electrolysis. The chapter only mentions electrolysis is used for extracting more reactive metals such as aluminium,

sodium, magnesium.

### *Question 19.* *[Combined Science 2025 21]*

**27** An electric current is passed through molten calcium chloride using graphite electrodes.

Which row shows the electrode products?

| | anode product | cathode product |
|---|---|---|
| **A** | calcium | chlorine |
| **B** | chlorine | calcium |
| **C** | chlorine | hydrogen |
| **D** | hydrogen | chlorine |

Final answer: B (anode: chlorine; cathode: calcium)

**1.** Molten calcium chloride ($CaCl_2$) is a molten binary ionic compound made of only two elements: a metal (calcium) and a non-metal (chlorine). Because it is molten, the $Ca^{2+}$ and $Cl^-$ ions are free to move, so it can be electrolysed. [from: Electrolysis of molten binary ionic compounds]

**2.** In an electrolytic cell the cathode is the negative electrode and the anode is the positive electrode. Graphite is an inert electrode, so it does not react with the electrolyte or the products. [from: Electrolytic cells and electrodes]

**3.** Positive ions (cations) move to the cathode and gain electrons (reduction); negative ions (anions) move to the anode and lose electrons (oxidation). So $Ca^{2+}$ moves to the cathode and $Cl^-$ moves to the anode. [from: Movement of ions and charge transfer]

**4.** For a molten binary ionic compound the metal is always formed at the cathode and the non-metal is always formed at the anode. Therefore calcium metal is produced at the cathode and chlorine gas is produced at the anode. [from: Electrolysis of molten binary ionic compounds]

**5.** The question asks for anode product / cathode product. That order is chlorine / calcium, which matches option B. [from: Electrolysis of molten binary ionic compounds]

From your chapter: Electrolysis of molten binary ionic compounds; Electrolytic cells and electrodes; Movement of ions and charge transfer; Binary compound; Inert electrode

### *Question 20.* *[Combined Science 2025 33]*

**5** Dilute sulfuric acid is used in an electrolysis experiment.

**(a)** Universal indicator is used to measure the pH of the dilute sulfuric acid.

Describe how to use universal indicator to measure pH.

.....

.....

.....

**(b)** Fig. 5.1 shows the electrolysis.

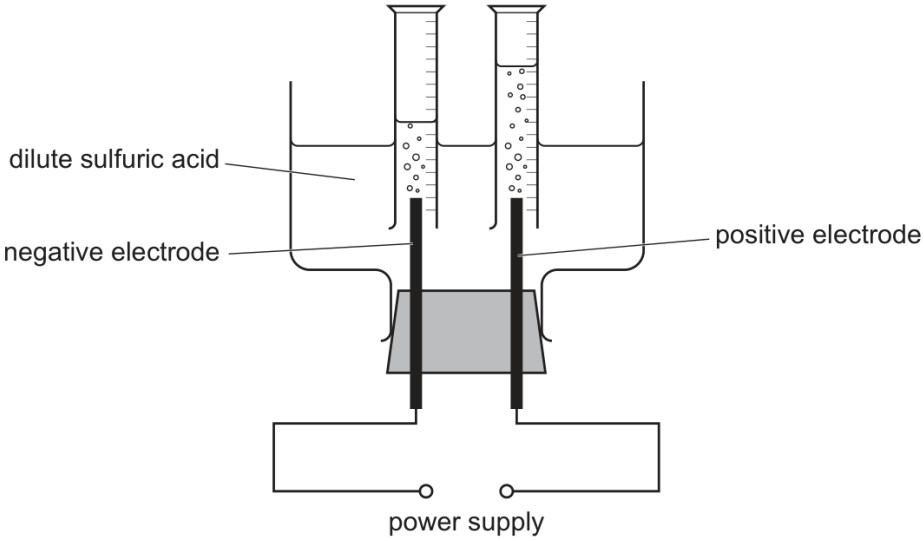



**Fig. 5.1**

**(i)** Identify the gas formed at each electrode.

negative electrode .....

positive electrode .....

**(ii)** State the name of the positive electrode.

.....

**(iii)** Identify the electrolyte in this electrolysis.

.....

**(c)** The volume of gas formed at each electrode is measured every 30 s for 2 min. The resu
shown in Table 5.1.

**Table 5.1**

| time / s | volume of gas at the negative electrode / $cm^3$ | volume of gas at the positive electrode / $cm^3$ |
|---|---|---|
| 0 | 0 | 0 |
| 30 | 18 | 9 |
| 60 | 36 | 18 |
| 90 | 54 | 27 |
| 120 | | 36 |

**(i)** Use Table 5.1 to predict the volume of gas given off at the negative electrode at 1

volume = ..... c

**(ii)** Use Table 5.1 to calculate the volume of gas formed per second at the positive ele

Final answer: (a) Dip universal indicator paper into the dilute sulfuric acid (or add a few drops of universal i

**1.** (a) Add a small amount of universal indicator (either a few drops of solution to a sample of the acid, or a piece of universal indicator paper dipped into the acid). The H+ ions in the dilute sulfuric acid make the indicator change colour. [from: Electrolyte]

**2.** (a) Compare the colour the indicator turns to the pH colour chart (the reference scale printed on the indicator bottle or paper) and read off the matching pH number. For dilute sulfuric acid, an acidic pH (around 1-2, red/orange) is expected. [from: Electrolyte]

**3.** (b)(i) Dilute H2SO4 contains H+, OH- and SO4^2- ions in solution. At the negative electrode (cathode), H+ ions are attracted there because opposite charges attract, and they gain electrons to form hydrogen gas: 4H+ + 4e- -> 2H2. [from: Electrolysis of dilute sulfuric acid]

**4.** (b)(i) At the positive electrode, OH- ions (from the water) lose electrons to form oxygen gas. The overall effect is that water is broken down: 2H2O(l) -> 2H2(g) + O2(g), giving hydrogen at the negative electrode and oxygen at the positive electrode in a 2:1 volume ratio. [from: Electrolysis of dilute sulfuric acid]

**5.** (b)(ii) The positive electrode in an electrolytic cell is always called the anode (the negative electrode is the cathode). [from: Electrolytic cells and electrodes]

**6.** (b)(iii) The electrolyte is the liquid that conducts electricity by the movement of ions and is decomposed in the process. In this experiment the liquid being electrolysed is the dilute sulfuric acid, so dilute sulfuric acid is the electrolyte. [from: Electrolyte]

From your chapter: Electrolyte; Electrolysis of dilute sulfuric acid; Electrolytic cells and electrodes; 4H+ + 4e- -> 2H2; 2H2O(l) -> 2H2(g) + O2(g)

## *Question 21.* *[Combined Science 2025 33]*

| / s | negative electrode / $cm^3$ | positive electrode / $cm^3$ |
|---|---|---|
| 0 | 0 | 0 |
| 30 | 18 | 9 |
| 60 | 36 | 18 |
| 90 | 54 | 27 |
| 120 | | 36 |

**(i)** Use Table 5.1 to predict the volume of gas given off at the negative electrode at 120 s.

volume = ..... $cm^3$ [1]

**(ii)** Use Table 5.1 to calculate the volume of gas formed per second at the positive electrode.

volume = ..... $cm^3/s$ [1]

[Total: 8]

Final answer: (c)(i) 72 cm^3; (c)(ii) 0.30 cm^3/s

**1.** (c)(i) Look at the volume of gas at the negative electrode at each 30 s interval in Table 5.1: 0, 18, 36, 54 cm^3 at 0, 30, 60, 90 s. The volume goes up by 18 cm^3 every 30 s, so gas is produced at the cathode (the negative electrode) at a steady rate. [from: Electrolysis of dilute sulfuric acid]

**2.** (c)(i) Continue the pattern one more 30 s step: volume at 120 s = 54 + 18 = 72 cm^3. This is the hydrogen produced at the cathode, where H+ ions gain electrons (4H+ + 4e- -> 2H2). [from: 4H+ + 4e- -> 2H2]

**3.** (c)(ii) Read the total volume of gas at the positive electrode (the anode) after 120 s from Table 5.1: 36 cm^3. This is the oxygen formed when OH-/O^2- lose electrons at the anode (2O^2- -> O2 + 4e-). [from: 2O^2- -> O2 + 4e-]

**4.** (c)(ii) Volume per second = total volume / total time = 36 cm^3 / 120 s = 0.30 cm^3/s. [from: Electrolysis of dilute sulfuric acid]

From your chapter: Electrolysis of dilute sulfuric acid; 4H+ + 4e- -> 2H2; 2O^2- -> O2 + 4e-

## *Question 22.* *[Combined Science 2025 32]*

**6** Table 6.1 shows the products and observations at the electrodes in the electrolysis of three electrolytes.

**Table 6.1**

| electrolyte | negative electrode | | positive electrode | |
|---|---|---|---|---|
| | product | observation | product | observation |
| molten lead(II) bromide | lead | grey liquid | ..... | bubbles of red-brown gas |
| concentrated aqueous sodium chloride | hydrogen | bubbles of colourless gas | chlorine | bubbles of green gas |
| dilute sulfuric acid | ..... | bubbles of colourless gas | ..... | bubbles of colourless gas |

**(a)** Complete Table 6.1. [3]

**(b)** Explain why the mass of concentrated aqueous sodium chloride decreases during electrolysis.

.....

..... [1]

**(c)** Use Table 6.1 to explain why electrolysis is a chemical change.

.....

..... [1]

Final answer: (a) Row 1 (molten lead(II) bromide) positive electrode: bromine. Row 3 (dilute sulfuric acid) n

**1.** (a) Row 1 - molten lead(II) bromide. Lead(II) bromide is a binary ionic compound (a compound made of just two elements, one metal and one non-metal). When it is melted, the metal forms at the cathode (the negative electrode) and the non-metal forms at the anode (the positive electrode). Lead is already shown at the negative electrode, so the non-metal bromine must form at the positive electrode. Bromine vapour is red-brown, which matches the observation 'bubbles of red-brown gas'. Answer for row 1, positive electrode: bromine. [from: Electrolysis of molten binary ionic compounds]

**2.** (a) Row 3 - dilute sulfuric acid. In dilute H2SO4 the ions present are H+, OH- and SO4^2-. Positive H+ ions move to the cathode (the negative electrode) and are discharged (gain electrons) to form hydrogen gas. Negative OH- ions (from the water) move to the anode (the positive electrode) and are discharged (lose electrons) to form oxygen gas. Both gases are colourless, which matches the table observations. [from: Electrolysis of dilute sulfuric acid]

**3.** (a) Therefore the three blanks in Table 6.1 are: Row 1 (molten lead(II) bromide), positive electrode product = bromine. Row 3 (dilute sulfuric acid), negative electrode product = hydrogen; positive electrode product = oxygen. [from: Electrolysis of molten binary ionic compounds]

**4.** (b) In concentrated aqueous sodium chloride the ions are Na+, H+, Cl- and OH-. H+ is discharged (gains electrons) at the cathode (the negative electrode) to form hydrogen gas. Cl- is discharged (loses electrons) at the anode (the positive electrode) to form chlorine gas. Both of these products are gases that bubble off and leave the solution, so the total mass of the electrolyte falls. [from: Electrolysis of concentrated sodium chloride solution]

**5.** (c) A chemical change produces new substances that are different from the starting materials. Table 6.1 shows this is exactly what happens: molten lead(II) bromide turns into lead and bromine; concentrated sodium chloride solution gives hydrogen and chlorine; dilute sulfuric acid gives hydrogen and oxygen. Because new substances are formed in every case, electrolysis is a chemical change. [from: Electrolysis: the breakdown of an ionic compound, when molten or in solution, by the passage of an electric current.]

**6.** (d) An inert electrode is an electrode that does not react with the electrolyte or with the products made during electrolysis. The chapter lists graphite and platinum as the standard inert electrode materials. Answer: graphite (platinum is also acceptable). [from: Inert electrode: an electrode that does not react with the electrolyte or the products (e.g. graphite, platinum).]

**7.** (e) In any electrolytic cell the electrode connected to the negative terminal of the battery is called the cathode. This is the electrode where positive ions (cations) gain electrons. Answer: cathode. [from: Cathode: the negative electrode in an electrolytic cell; site of reduction; cations are discharged here.]

**8.** (f) Electrolysis only happens when the battery supplies electrical energy that is taken in by the electrolyte to break apart the ionic compound. A process that takes in (absorbs) energy from its surroundings is called endothermic. Answer: endothermic means a process that takes in / absorbs energy from the surroundings. [from: Electrolysis: the breakdown of an ionic compound, when molten or in solution, by the passage of an electric current.]

From your chapter: Electrolysis of molten binary ionic compounds; Electrolysis of concentrated sodium chloride solution; Electrolysis of dilute sulfuric acid; Electrolysis: the breakdown of an ionic compound, when molten or in solution, by the passage of an electric current.; Inert electrode: an electrode that does not react with the electrolyte or the products (e.g. graphite, platinum).; Cathode: the negative electrode in an electrolytic cell; site of reduction; cations are discharged here.; Movement of ions and charge transfer

## *Question 23.* *[Combined Science 2025 32]*

**(g)** Draw **one** straight line from each description to the correct substance.

| **description** | **substance** |
|---|---|
| | chlorine |
| salt that contains ionic bonds | |
| | dilute sulfuric acid |
| compound with a pH less than 7 | |
| | hydrogen |
| | lead |
| Group VII element | |
| | lead(II) bromide |

[3]

[Total: 11]

Final answer: salt with ionic bonds -> lead(II) bromide; compound with pH < 7 -> dilute sulfuric acid; Group

**1.** (g) Identify which substance is a salt held together by ionic bonds. Lead(II) bromide (PbBr2) is a binary ionic compound made of Pb^2+ and Br- ions. Because it is made of oppositely charged ions held together by ionic bonding, it is the only 'salt with ionic bonds' in the list. Lead is a pure metal (metallic bonding, not ionic), hydrogen and chlorine are non-metal molecules (covalent bonding), and dilute sulfuric acid is an acid in water, not a solid salt. So the first line goes from 'salt that contains ionic bonds' to lead(II) bromide. [from: Electrolysis of molten binary ionic compounds]

**2.** (g) Identify which substance is a compound with a pH less than 7. A pH below 7 means the substance is acidic, which happens when it releases H+ ions in water. From the chapter, dilute sulfuric acid contains H+, OH- and SO4^2- ions in solution, and the H+ ions are what make it acidic. None of the other substances release H+ in water (lead(II) bromide gives Pb^2+ and Br-, lead is a metal, chlorine and hydrogen are elements, not compounds). So the second line goes from 'compound with a pH less than 7' to dilute sulfuric acid. [from: Electrolysis of dilute sulfuric acid]

**3.** (g) Identify which substance is a Group VII element. Group VII of the Periodic Table is the halogen family, and chlorine (Cl2) is a halogen. The chapter treats chlorine as a non-metal element produced at the anode when concentrated sodium chloride solution is electrolysed, which matches the fact that it is a Group VII element. Hydrogen sits in Group I (or is shown separately), lead is a metal, and lead(II) bromide and dilute sulfuric acid are compounds not elements. So the third line goes from 'Group VII element' to chlorine. [from: Electrolysis of concentrated sodium chloride solution]

From your chapter: Electrolysis of molten binary ionic compounds; Electrolysis of dilute sulfuric acid; Electrolysis of concentrated sodium chloride solution; Binary compound: a compound containing only two different elements (e.g.

PbBr2, NaCl).; ionic bonding

Generated 2026-04-25T21:20:30+00:00